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A Candidate Third Transiting Planet on an 18.9-day Orbit around the M Dwarf TOI-4342, Matching a Dynamical Prediction

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ABSTRACT

I present an independent detection and detailed vetting of a candidate third transiting planet around TOI-4342 (TIC 354944123), an M0 dwarf hosting two sub-Neptunes at 5.54 and 10.69 days; the signal was first posted publicly by Eumemic (2026). Nine sectors of TESS 2-minute photometry show a 1182 ± 107 ppm, 2.4 hr transit every 18.908226 days, $R_p = 2.25 \pm 0.13\,R_\oplus$, in all ten observed windows (signal-to-noise ratio 11.7). Period, radius and inclination match the planet predicted by E. V. Turtelboom et al. (2025). Difference images place the source within 11″ of the target and exclude every Gaia neighbor able to mimic it. With a speckle contrast curve and the pixel-excluded neighbors removed, TRICERATOPS gives a false-positive probability of 0.018, just above the 0.015 validation threshold, and a nearby-false-positive probability of zero. The three planets are coplanar to within 1° and dynamically stable.

Keywords: Exoplanet detection methods (489) — Exoplanets (498) — M dwarf stars (982) — Transit photometry (1709)

1. INTRODUCTION

TOI-4342 ($T = 11.03$, 61.5 pc; $T_{\rm eff} = 3866 \pm 92$ K, $R_\star = 0.598 \pm 0.018\,R_\odot$, $M_\star = 0.587 \pm 0.013\,M_\odot$) hosts planets b (5.538 d) and c (10.689 d) near the 2:1 resonance (E. Tey et al. 2023), with masses 7.3 ± 1.3 and $4.8 \pm 1.4\,M_\oplus$ (L. Parc et al. 2026). E. V. Turtelboom et al. (2025) predicted a further planet at $18.5^{+2.7}_{-2.0}$ d with $2.24\,R_\oplus$. This signal was first reported publicly on 2026 October 7 by Eumemic (2026), whose multi-sector TESS search lists it at 18.90820 d and $2.21 \pm 0.11\,R_\oplus$ with a TESS-only false-positive probability of 0.14. I detected it independently while searching small-star planet hosts for additional transits, and add difference-image localization, archival-plate and speckle limits, a speckle-constrained false-positive probability, per-transit depths and timing, and dynamics.

2. DETECTION

TESS (G. R. Ricker et al. 2015) observed TOI-4342 in Sectors 13, 27, 66, 67, 93, 94 and 101–103 (2019–2026). I used SPOC 2-minute light curves and target pixel files (J. M. Jenkins et al. 2016), 20-second data, and QLP and TESS-SPOC full-frame-image light curves, detrended with a 0.75-day biweight filter (transits masked, b and c subtracted). A box least-squares search found the signal.

Ten of fourteen predicted transits have data (one partial, one in cadences that SPOC's planet search excludes), and all ten show the transit (Figure 1b): depths agree ($\chi^2 = 8.7$, 9 degrees of freedom), as do the 2019–2023 and 2025–2026 halves (0.98 ± 0.11 and 0.89 ± 0.12 of the model depth). The signal appears in simple-aperture flux (10.7σ) and QLP (8.8σ). With the real transits removed, the same statistic at 3,000 random periods never exceeded 8.9 (observed: 13.4). Depths at $P/2$, $P/3$ and $P/4$ are consistent with zero (53 ± 78 ppm at $P/2$); odd and even transits agree (0.5σ), excluding $2P$. Transit times fit a linear ephemeris ($\chi^2 = 7.9$, 6 degrees of freedom).

A `batman/emcee` fit with a stellar-density prior of 3.86 ± 0.36 g cm^{-3} gives $P = 18.908226 \pm 0.000042$ d, $T_0 = 2459904.5666 \pm 0.0021$ BJD$_{\rm TDB}$, $R_p/R_\star = 0.0344 \pm 0.0016$, $b = 0.76 \pm 0.03$, $i = 88.96 \pm 0.07°$, $a/R_\star = 41.7 \pm 1.3$ and duration 2.44 ± 0.11 hr (Figure 1a). Fitted freely, the density ($8.8^{+4.7}_{-5.9}$ g cm^{-3}) is consistent with the star. The inclination matches b ($88.83 \pm 0.22°$) and c ($89.61 \pm 0.26°$).

3. VETTING

Gaia DR3 (Gaia Collaboration et al. 2016, 2023) RUWE is 0.98. Gemini-South/Zorro speckle imaging (E. Tey et al. 2023) detected no companion at $\Delta m_{832} > 4.7$ beyond $0.1''$. A 1976 photographic plate, when the star was $7.4''$ from its 2026 position, shows no source there to $\Delta T \approx 7$; a 1182 ppm signal requires $\Delta T < 7.4$.

I fitted each sector's in-transit difference image with its out-of-transit image shifted across a sky grid, summing χ^2 over sectors. The candidate gives a target $\Delta\chi^2$ of 11.4 (b and c: 9.5 and 3.5 on the same pixels), a 3σ region within $11''$ of the target, and $\Delta\chi^2 \geq 79$ for every Gaia star that could mimic it (Figure 1c). Its depth versus aperture size tracks those of b and c.

TRICERATOPS (S. Giacalone et al. 2021), with nine sectors' SPOC apertures and the Zorro contrast curve, gives a false-positive probability (FPP) of 0.0161 ± 0.0002 and a nearby-false-positive probability (NFPP) of 0.0005. Removing the pixel-excluded neighbors gives FPP $= 0.018 \pm 0.002$ and NFPP $= 0$; adding MOLUSC (M. L. Wood et al. 2021) bound-companion limits gives 0.016 ± 0.011, against 0.14 from TESS alone (Eumemic 2026). The remainder is almost entirely a planet transiting an unresolved bound companion. The candidate thus sits at, not below, the validation threshold (FPP < 0.015, NFPP < 0.001; S. Giacalone et al. 2021), and I do not claim validation. I apply no Kepler-calibrated multiplicity boost, which would lower the FPP.

N-body models predict transit-timing variations of a few minutes, below the ≈8-minute per-transit precision; the observed times neither require nor exclude them. SPOCK (D. Tamayo et al. 2020) gives a stability probability of 0.89 (median over mass and eccentricity draws; 97% of draws above 0.5) for b, c, the candidate and the 47.5-day radial-velocity candidate of L. Parc et al. (2026).

4. A SECOND, WEAKER SIGNAL

A 287 ± 40 ppm transit also recurs every 1.957690 days ($1.11\,R_\oplus$, signal-to-noise ratio 8.8), consistent across seasons, with no odd–even difference and the expected density; Eumemic (2026) list it, unpromoted, among their search signals. Its localization is looser; TRICERATOPS gives FPP = 0.013 (0.007 with MOLUSC) and NFPP = 0.001, but its signal-to-noise ratio is near my search's expected noise-detection level, so it is a lower-confidence candidate.

5. FOLLOW-UP, DATA AND CODE

TESS will not observe TOI-4342 again through Sector 134; the 1.2 ppt, 2.4 hr transit is within reach of 1-m-class telescopes. Predicted mid-transit times are $T_n = 2459904.5666 + 18.908226\,n$ $\mathrm{BJD_{TDB}}$ (± 12 minutes through 2027), including fully observable southern transits on 2026 October 28 02:10 (Chile), 2027 May 24 01:56 (South Africa) and 2027 August 26 14:55 UTC (Australia). TESS data: MAST, doi:10.17909/t9-nmc8-f686 and doi:10.17909/t9-wpz1-8s54. Code: https://github.com/rpinho/planet-hunt (branch `round6-toi4342`).

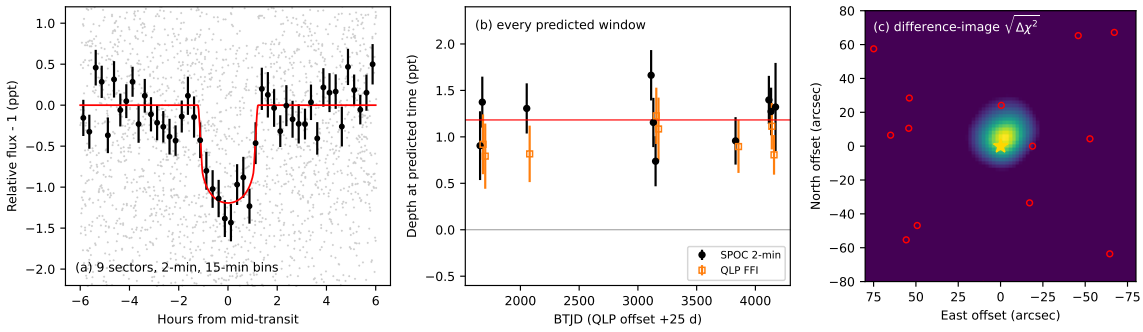


Figure 1. (a) Nine 2-minute sectors, with b, c and the 1.96-day signal removed, folded on the 18.9-day ephemeris (15-minute bins), with the best-fit model. (b) Depth at each predicted transit, 2-minute (filled) and QLP (open, offset 25 days); line: model depth. (c) Difference-image $\sqrt{\Delta\chi^2}$ over nine sectors; star: TOI-4342; circles: Gaia stars able to mimic the signal.

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AI DISCLOSURE STATEMENT

Generative AI was used extensively. Claude Opus 5.5 (Anthropic) wrote and ran the analysis code, carried out the data retrieval, vetting, statistical and dynamical tests, searched the literature, and drafted the text, figure and bibliography.

Facilities: TESS, Gaia, Gemini:South (Zorro), ExoFOP

Software: astropy, batman, emcee, lightkurve, matplotlib, MOLUSC, numpy, REBOUND, scipy, SPOCK, tesspoint, TRICERATOPS, wotan

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